Exchangeability and the Dirichlet Limit of a Multicolor Pólya Urn

Route-A source-local certificate HCS-C263

31 August 2026

Abstract

We close the classical multicolor Pólya urn from ordered histories through all factorial moments to its Dirichlet directing measure. Finite-time exchangeability and the de Finetti mixture are proved in both directions, and the normalized masses converge almost surely and in every finite L^p. Zero reinforcement, zero initial masses, and one color are retained without undefined normalization. No target arithmetic or spectral data are used.

1 Frozen urn

This is the classical reinforced scheme of Eggenberger and Pólya [1]. Let $a_i \geq 0$, $\sum_i a_i > 0$, and first suppose $c > 0$. A zero-mass color is never drawn, so delete it and put $\alpha_i = a_i/c > 0$ and $A = \sum_i \alpha_i$ on the active face. At each integer time a color is drawn proportionally to its current mass, replaced, and receives additional mass c. Write $N_i(n)$ for its count among the first n draws and use the rising factorial $(x)^{\overline{r}} = x(x+1)\cdots(x+r-1)$.

Theorem 1 (word and count law)**.** *If an ordered word $w = (w_1, \ldots, w_n)$ has multiplicities n_i, then*

$$\mathbb{P}(w) = \frac{\prod_{i=1}^{K} (\alpha_i)^{\overline{n_i}}}{(A)^{\overline{n}}}. \tag{1}$$

Consequently

$$\mathbb{P}\{N(n) = (n_1, \ldots, n_K)\} = \frac{n!}{\prod_i n_i!} \frac{\prod_i (\alpha_i)^{\overline{n_i}}}{(A)^{\overline{n}}}. \tag{2}$$

Proof. Before draw $t+1$, the predictive probability of color i is $(\alpha_i + N_i(t))/(A+t)$. Multiplication along a word groups the numerator factors by color and yields (1), independent of order. There are $n!/\prod_i n_i!$ words with the given counts, proving (2). Summing (2) gives one by the multivariate rising-factorial Vandermonde identity. □

2 Marginals and every factorial moment

Write $(x)_{[r]} = x(x-1)\cdots(x-r+1)$ for a falling factorial. The distinction from the rising factorial in (1)–(2) is essential.

Theorem 2 (moment closure)**.** *Every color count is beta–binomial:*

$$\mathbb{P}\{N_i(n) = x\} = \binom{n}{x} \frac{(\alpha_i)^{\overline{x}} (A - \alpha_i)^{\overline{n-x}}}{(A)^{\overline{n}}}.$$

For a multi-index $r = (r_1, \ldots, r_K)$ and $R = \sum_i r_i$,

$$\mathbb{E} \prod_i (N_i(n))_{[r_i]} = (n)_{[R]} \frac{\prod_i (\alpha_i)^{\overline{r_i}}}{(A)^{\overline{R}}}. \tag{3}$$

Hence

$$\begin{aligned} \mathbb{E} N_i &= \frac{n\alpha_i}{A}, \\ \operatorname{Var} N_i &= \frac{n\alpha_i (A - \alpha_i)(A+n)}{A^2(A+1)}, \\ \operatorname{Cov}(N_i, N_j) &= -\frac{n\alpha_i\alpha_j(A+n)}{A^2(A+1)} \quad (i \neq j). \end{aligned}$$

Proof. Sum (2) over the other colors and apply rising-factorial Vandermonde to obtain the marginal. To prove (3), mark an ordered selection of R draw positions, assigning r_i positions to color i. Exchangeability makes the probability of that assignment $\prod_i (\alpha_i)^{\overline{r_i}}/(A)^{\overline{R}}$; there are $(n)_{[R]}$ ordered marked positions. The first and second choices of r give the displayed mean and covariance formulas. ☐

3 Dirichlet mixture, converse, and limit

Let Θ have the Dirichlet distribution with active parameter vector α. Its monomial moments are

$$\mathbb{E}\prod_i \Theta_i^{n_i} = \frac{\prod_i (\alpha_i)^{\overline{n_i}}}{(A)^{\overline{n}}}. \tag{4}$$

Thus iid categorical draws conditional on Θ have exactly the ordered word law (1). Conversely, after a word with counts $N_i(n)$, Bayes' formula gives the posterior Dirichlet$(\alpha + N(n))$ and hence predictive probability $(\alpha_i + N_i(n))/(A+n)$, exactly the urn update. This proves the finite-time mixture in both directions, not merely equality of the first two moments; it is the finite-color specialization of the Pólya scheme discussed by Blackwell and MacQueen [2].

Define

$$P_i(n) = \frac{\alpha_i + N_i(n)}{A+n}.$$

Conditioning on the first n draws gives

$$\begin{aligned}\mathbb{E}[P_i(n+1) \mid \mathcal{F}_n] &= P_i(n)\frac{\alpha_i + N_i(n) + 1}{A+n+1} + (1-P_i(n))\frac{\alpha_i + N_i(n)}{A+n+1} \\ &= P_i(n).\end{aligned}$$

Thus each coordinate is a bounded martingale. On the mixture probability space, the conditional strong law gives $N(n)/n \to \Theta$ almost surely, while $P(n) - N(n)/n \to 0$. Therefore

$$P(n) \longrightarrow \Theta \sim \text{Dirichlet}(\alpha) \quad \text{almost surely.}$$

Every coordinate is bounded by one, so dominated convergence gives convergence in L^p for every finite p. This identifies the martingale limit, rather than only proving that a limit exists.

4 Boundary atlas and object separation

If $c = 0$, the masses do not change. The draws are iid with probabilities $p_i = a_i/\sum_j a_j$, $N(n)$ is multinomial, and no $\alpha = a/c$ is defined. The physical mass proportion stays equal to p, whereas the empirical count proportion $N(n)/n$ converges to p. If $a_i = 0$, color i remains absent and all positive-reinforcement formulas apply after deleting it. If $K = 1$, the unique word has probability one and every normalized proportion is one.

object	finite-time role	limiting role
$N_i(n)$	integer draw count	$N_i(n)/n \to \Theta_i$
$P_i(n)$	normalized current mass	bounded martingale $\to \Theta_i$
Θ_i	Dirichlet mixture coordinate	random directing probability

5 Certificate and Route-A boundary

The exact implementation freezes twelve cases and enumerates 10,860 ordered words, 1,000 composition rows, 1,021 marginal rows, 1,567 factorial-moment rows, and 2,042 conditional martingale rows. An independent implementation closes 62,233 assertions, SymPy closes 156 identities, byte replay is exact, and repaired/stale-hash mutation testing rejects 23/23 changes. Finite enumeration is a regression oracle; the proof above holds for every K, time, and admissible parameter.

Total mass grows from $\sum_i a_i$ to $\sum_i a_i + cn$. Therefore this owner has no nontrivial periodic source orbit, primitive repetition clock, or source Artin–Mazur product. It supplies no arithmetic local datum, Euler factor, root number, automorphy statement, target divisor, functional equation, or Hilbert–Pólya operator. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the strict tuple is

`(A0_FAIL,A1_FAIL,A2_FAIL,A3_FAIL,A4_FAIL).`

The verdict is `ROUTE_A_REJECTED`; Route B is disabled. The Dirichlet directing measure is a source probability representation, not a target spectral operator.

References

[1] F. Eggenberger and G. Pólya, *Über die Statistik verketteter Vorgänge*, *Z. Angew. Math. Mech.* **3** (1923), 279–289, doi:10.1002/zamm.19230030407.

[2] D. Blackwell and J. B. MacQueen, *Ferguson distributions via Pólya urn schemes*, *Ann. Statist.* **1** (1973), 353–355, doi:10.1214/aos/1176342372.